

HealthConnect Model Improvement & Validation Report

AnalystLab Africa Experience Lab - Week 6 | Data Science Track

1. Executive Summary

Week 6 focused on improving and validating the Week 5 no-show prediction model. The Week 5 Logistic Regression baseline achieved about **63% accuracy, 65% no-show recall, and 0.68 ROC-AUC**.

This week, I performed targeted feature analysis, investigated model errors, refined features, and compared alternative models. **Gradient Boosting** produced the strongest overall classification results, with about **65% accuracy, 65% no-show recall, 65% F1-score, and 0.68 ROC-AUC**. The improvement was modest, but it reduced false positives while maintaining similar no-show detection.

2. Targeted Feature Analysis

Previous No-Show History

Previous No-Shows	Current No-Show Rate
0	46.3%
1	55.9%
2	62.1%
3	69.7%

No-show rates generally increased as previous no-show history increased. This supported retaining previous attendance behaviour as an important modelling signal.

Reminder Status

Appointments without reminders had a **54.6%** no-show rate, compared with **49.9%** when reminders were sent. This is an observed association and does not prove a causal reminder effect.

Appointment Type and Reminders

The reminder association differed across appointment types. Diagnostic Tests showed the clearest difference: **61.1% no-show without a reminder** versus **49.3% with a reminder**. This supported testing interaction features during model refinement.

3. Baseline Error Analysis

Result	Count
True Negative	292
False Positive	191
False Negative	168
True Positive	315

Booking lead time strongly influenced how the baseline classified appointments. Correctly predicted no-shows averaged about **44 days** of booking lead time, while false positives averaged about **39 days**. False negatives averaged only about **18 days**. A similar pattern appeared for previous no-show history.

The baseline was therefore better at recognising clear historical risk patterns, but it struggled with less obvious no-shows and some high-risk-looking patients who still attended.

4. Feature Refinement

Feature engineering was guided by targeted analysis, cross-track Data Analytics findings, and baseline errors. Interaction features were tested for appointment type and reminder status, previous no-show history and reminders, and

booking lead time with previous no-show history.

The refined Logistic Regression showed a small improvement in classification performance, but ROC-AUC remained about **0.68**. The new features added limited new discrimination.

5. Model Comparison

Model	Accuracy	No-Show Recall	No-Show F1	ROC-AUC
Week 5 Logistic Regression	0.63	0.65	0.64	0.68
Refined Logistic Regression	0.64	0.66	0.64	0.68
Random Forest	0.61	0.62	0.61	0.64
Gradient Boosting	0.65	0.65	0.65	0.68

Random Forest did not improve performance, showing that greater model complexity did not automatically improve generalisation. Gradient Boosting produced the strongest overall classification performance.

6. Candidate Model

Gradient Boosting was selected as the Week 6 candidate model.

	Predicted Attended	Predicted No-Show
Actual Attended	308	175
Actual No-Show	167	316

Compared with the Week 5 baseline, Gradient Boosting reduced false positives from **191 to 175** while maintaining similar no-show detection. The improvement is useful but modest, so the model should continue to be validated rather than treated as final.

7. Threshold Analysis

At the default threshold of **0.50**, the model provided balanced performance. At a threshold of **0.45**, no-show recall increased to about **74%**, with about **60% precision** and **62% accuracy**.

A lower threshold may be useful when HealthConnect prioritises identifying more potential no-shows and the intervention is low-cost, such as an additional reminder. The final threshold should depend on operational capacity and the cost of false alerts.

8. Feature Interpretation

Random Forest feature importance ranked **booking lead time** highest. Distance to clinic, age, waiting time, previous appointment history, and the booking lead/history interaction also contributed. These importance values were treated as model-specific because Random Forest itself performed worse than the selected candidate model.

9. Cross-Track Integration

Data Analytics findings guided deeper analysis and feature refinement. Findings around previous no-show history, reminders, booking lead time, appointment type, and distance were compared with the Data Science results and used to identify relationships worth testing.

The final candidate model and its preprocessing requirements are also being prepared for handoff to the ML Engineering track for pipeline integration. This turns cross-track findings into concrete modelling decisions and integration inputs.

10. Limitations

Model performance remains moderate. ROC-AUC did not improve beyond about **0.68**. Some no-shows remain difficult to identify, especially patients without strong historical risk patterns. The dataset is fictional, so results cannot automatically be generalised to real clinical settings. Reminder findings are associations and should not be interpreted as causal effects.

11. Week 7 Testing Requirements

Week 7 should validate Gradient Boosting on consistent patient-level splits, test the proposed decision threshold, examine false negatives further, validate preprocessing and model integration, test stability and reproducibility, review fairness across relevant groups, and confirm that outputs are suitable for the intended administrative use.

Conclusion

Week 6 improved the HealthConnect modelling workflow through targeted analysis, error investigation, feature refinement, model comparison, and threshold validation. Gradient Boosting was selected as the candidate model because it provided the strongest overall classification performance. It now provides a better-validated foundation for Week 7 testing.